

Neuromarketing Study: McDonald's

How Emotion vs. Product Focus
Drives Attention & Purchase Intent

Course: MKT 546 - Neuromarketing

Professor: Dr. Jennifer Tatara

Group 7

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Agenda / Project Overview



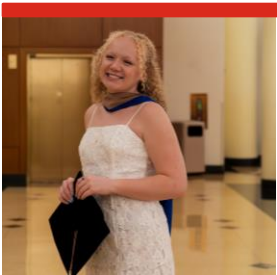
AGENDA

01 Background & objectives
02 Hypothesis
03 Stimulus A & B
04 Our sample
05 Study design, methodology & data collection
06 Study results
07 Advertising effectiveness & CX implications
08 Limitations & future directions
09 Conclusion
10 Appendix

PROJECT AT A GLANCE

Brand	McDonald's - Big Mac digital ad campaign
Tools	iMotions (Eye Tracking) · Qualtrics (Survey)
Target	Adults 18–35, Chicago, IL
Sample	N = 6 participants · n = 3 per stimuli
Stimuli	Version A: Product-forward · Version B: Emotion-forward
Key AOIs	Logo · Headline · Product / Face · Social Proof · CTA

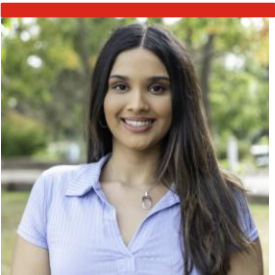
MEET THE TEAM



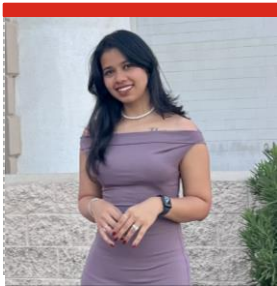
Francesca



Disha



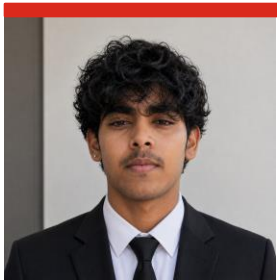
Kristal



Khushi



Wafa



Naveen

Background



McDONALD'S

Founded

1940

San Bernardino, CA

"i'm lovin' it"

Global campaign since 2003

Market Cap

\$220-240B

Record high: Feb 2026

Company Overview

Fast food's most recognizable global brand

40,000+

Restaurants Worldwide

in 100+ countries

\$25.5B

Annual Revenue

2023 figure

95%+

Franchised Locations

franchise-operated

\$30B+

Digital Sales 2025

~20% of system revenue

2,276

New Units Opened 2025

2,600 planned for 2026

\$140B+

Systemwide Sales

Global sales across all restaurants

Business model: 95%+ franchised · Owns \$37.7B in real estate · Target: 50,000 total locations

RESEARCH OBJECTIVE

Objective

FOCUS

An eye-tracking and consumer-neuroscience study of emotion-led vs product-led McDonald's advertising.

THE GOAL

Using eye-tracking and consumer neuroscience to understand how emotionally driven vs. product-focused McDonald's advertisements influence:

- Visual attention
- Emotional engagement
- Purchase intention



WHAT WE MEASURE

Analyze consumer attention using:

- AOIs (Areas of Interest)
- Fixation duration
- Dwell time
- Time to first fixation



WHAT WE EXPECT

We expect Version A to perform better because:

- Product-focused layout creates stronger visual salience and consumption cues
- Faster System 1 decision-making: Big Mac image bypasses rational processing
- Version B's face creates cognitive load, diverting attention from product & CTA



WHAT WE PREDICTED

Hypothesis

H1

Version A (product-focused) is expected to outperform Version B (emotion-focused).

VERSION A - PREDICTED

Participants exposed to Version A are expected to demonstrate:

- Greater fixation duration on the burger (product) AOI
- Higher dwell time on the product image and CTA
- Stronger overall visual attention
- Higher purchase intention



VERSION B - PREDICTED

Participants exposed to Version B may show:

- Greater emotional engagement toward the facial AOI
- Increased cognitive load from emotional imagery
- Attention diverted away from product and logo AOIs
- Lower immediate purchase intent



NET PREDICTION

Therefore, Version A is expected to be more effective in driving advertising effectiveness and purchase intent:

- Product-focused layout creates stronger visual salience and consumption cues
- Faster System 1 decision-making via clear brand-product association



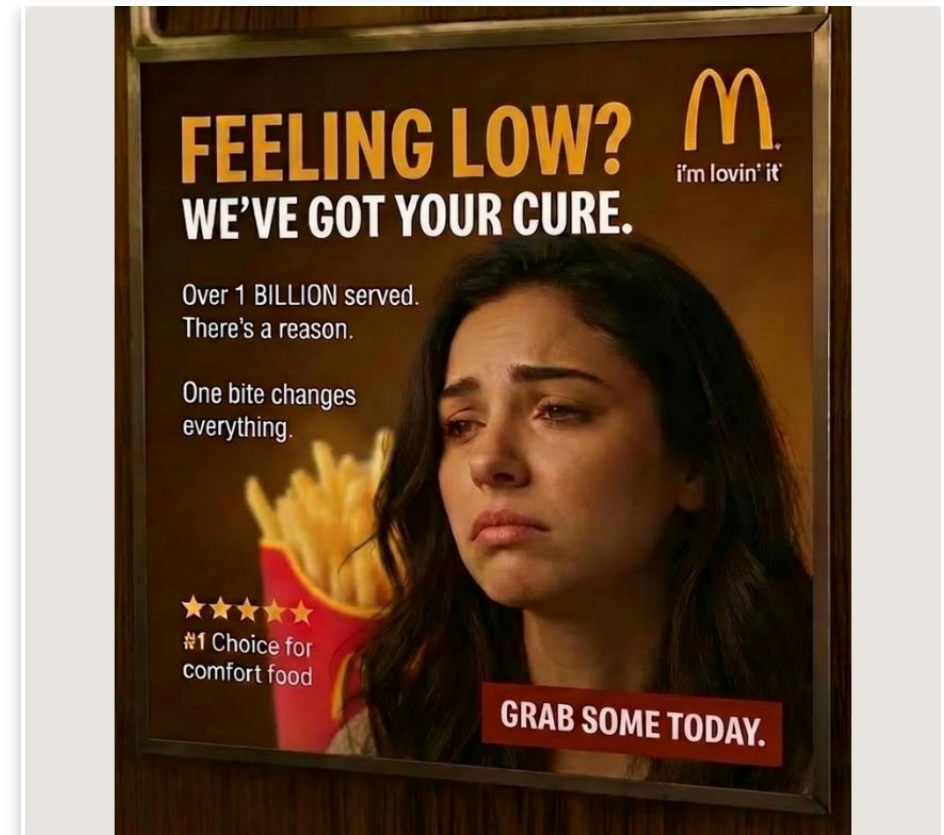
Stimuli Comparison

Two ad concepts tested · A vs B

VERSION A - Product-Focused



VERSION B - Emotion-Focused



Areas of Interest

5

Areas of Interest (AOIs)

- 01 Logo
- 02 Headline
- 03 Product / Face
- 04 Social Proof (Rating)
- 05 CTA (Call to Action)

VERSION A - PRODUCT-FOCUSED



VERSION B - EMOTION-FOCUSED

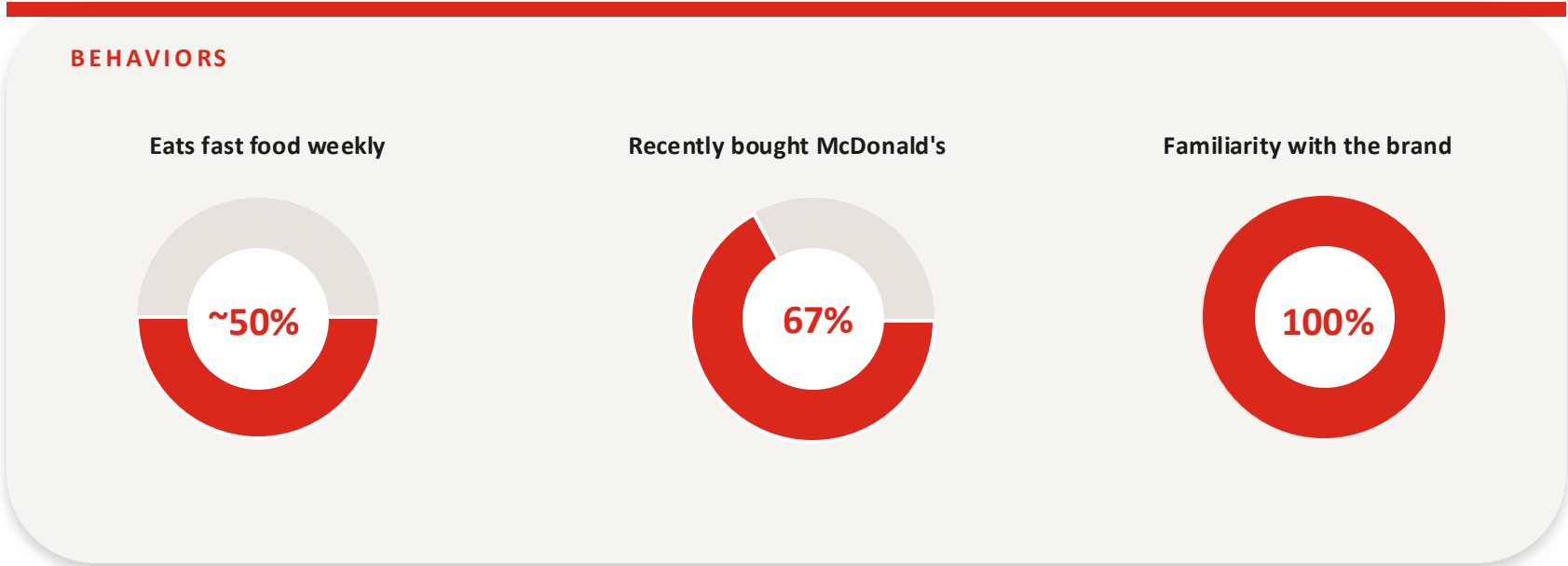
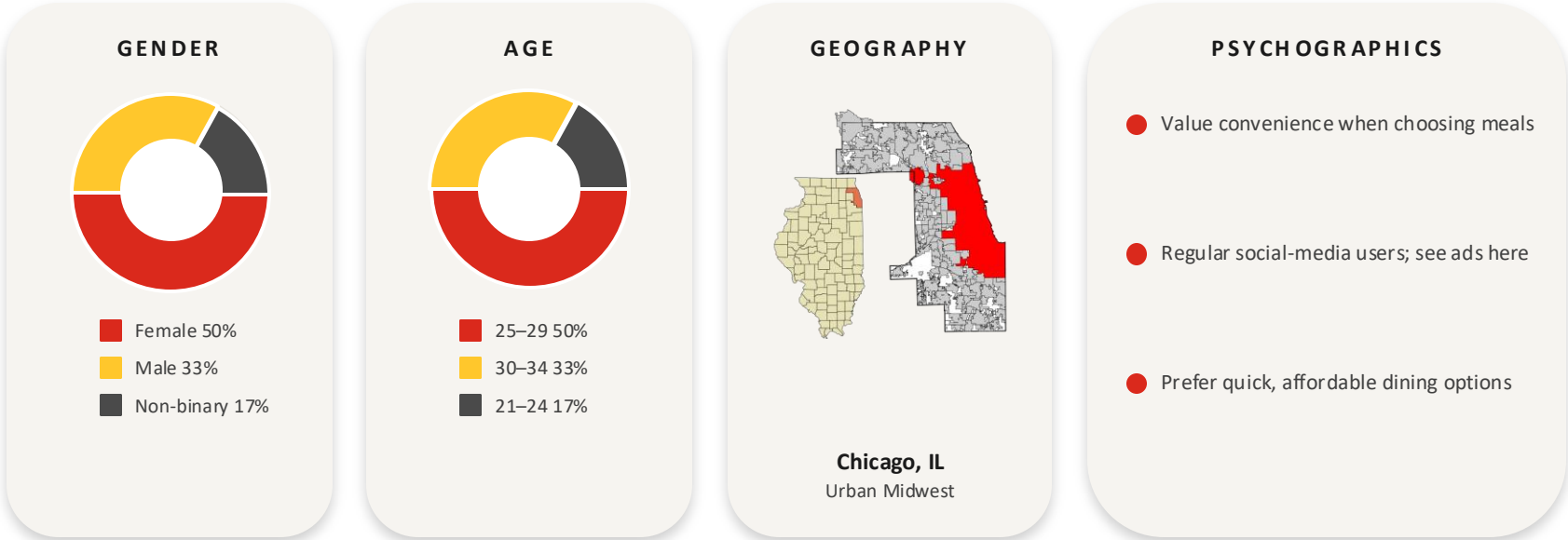


WHO WE STUDIED

About Our Sample

SNAPSHOT

N = 6 participants · n = 3 per version
· ages 18–35 · 100% familiar with McDonald's



01

Eye-Tracking Study

- Consent collected
- Viewed McDonald's ad Version A or B
- Eye movements tracked via iMotions



02

Post-Study Survey

- Follow-up questions via Qualtrics
- Behavioral & demographic items
- Lifestyle (psychographic) items
- Advertisement-focused items



03

Data Analysis

- iMotions heatmaps
- Bee-swarm gaze plots
- AOI metrics
- Survey responses



04

Conclusions

- Evaluated ad effectiveness
- Compared consumer attention
- Compared purchase intent across both ads



Bee Swarm

Product-Focused

WHAT YOU'RE SEEING

Each colored dot marks one participant's fixation. Denser, warmer clusters mean longer and more frequent looking.



Version A - "BAD DAY? BITE INTO BETTER." (product-focused)

WHAT THE PLOT SHOWS

Fixations spread across the Big Mac, headline and CTA - product cues hold attention and support recall.

Bee Swarm

Emotion-Focused

WHAT YOU'RE SEEING

Each colored dot marks one participant's fixation. Denser, warmer clusters mean longer and more frequent looking.



Version B - "FEELING LOW? WE'VE GOT YOUR CURE." (emotion-focused)

WHAT THE PLOT SHOWS

Gaze locks onto the face first (TTFF $\approx$ 0 ms) - the Vampire Effect crowds out the product, logo and CTA.

Heatmaps Comparison

Gaze intensity · A vs B

VERSION A - Product-Focused



VERSION B - Emotion-Focused



Hence, version A created more balance viewing pattern

WHAT THE GAZE DATA SHOWS

Eye Tracking Results



KEY INSIGHT

Emotion captures attention instantly - but product-focused layouts drive stronger recall, brand clarity and purchase intent.

VERSION A - PRODUCT-FOCUSED

- Balanced gaze across Big Mac, CTA & social proof
- 67% Big Mac recall after exposure
- Higher social-proof fixations (6.3 vs 1.0)

VERSION B - EMOTION-FOCUSED

- Face captured gaze instantly (TTFF = 0 ms) - Vampire Effect
- Higher headline dwell (1,700 vs 972 ms) - only sig. result
- 0% Big Mac recall - face dominated encoding

AOI METRICS · A vs B

	VER A	VER B
Product (Big Mac) recall	67%	0%
Logo noticeability	100%	67%
Headline dwell	972 ms	1,700 ms
Social-proof fixations	6.3	1.0
Ad clarity (1–10)	6.0	4.7
Purchase intent (1–5)	2.0	1.3

ANOVA Results



SAMPLE NOTE

$N \geq 15$ per cell is needed for inference. At $N = 6$ ($n = 3/\text{cell}$) results are directional.

p = .035

Headline Dwell - SIGNIFICANT

$F(1,4) = 9.95$ · Version A 972 ms vs Version B 1,700 ms · partially supports H1

p = .137

Social Proof Dwell - n.s.

$F(1,3) = 4.06$ · A 1,684 ms vs B 508 ms · large effect, not significant

p = .563

Purchase Intent - n.s.

$F(1,4) = 0.40$ · A 2.0 vs B 1.3 · not significant

p = .651

Ad Clarity - n.s.

$F(1,4) = 0.24$ · A 6.0 vs B 4.7 · not significant

STRATEGIC IMPLICATIONS

Advertising Effectiveness



CORE FINDING

Emotion captures attention - but product clarity drives recall, brand recognition, and purchase intent. A hybrid approach optimizes both.

The Hybrid Approach

01

Combine Version B's emotional headline with Version A's product-focused layout - emotional engagement while maintaining product clarity and brand recall.

Minimize Face Dominance

02

Limit expressive faces that swallow attention from logo and product. The Vampire Effect (TTFF = 0 ms on face) proves faces compete with brand assets.

Product at Eye Level

03

Position the Big Mac at eye level to trigger instant decision-making. Product-forward layout creates stronger visual salience and faster System 1 processing.

CX IMPLICATIONS

Customer Experience Implication

Data-driven ways to improve McDonald's digital touchpoints, reduce cognitive load, and lift purchase intent.

Simplify Digital Menus

01

Reduce textual complexity in digital menus. Visual cues should guide the customer's hunger, not their logic - fewer words, stronger imagery.

"#1 Choice" Social-Proof Badges

02

Integrate "#1 Choice" badges into digital touchpoints. Our data shows these ratings capture high visual engagement and reinforce purchase confidence.

Align CTA with Natural Gaze

03

Place the Call to Action along natural gaze patterns (bottom-right) to increase purchase intent and reduce friction in the decision journey.

WHAT CONSTRAINS THE FINDINGS

Study Limitations



STATISTICAL NOTE

$N \geq 15$ per cell is required for inferential conclusions. Current $N = 6$ ($n = 3$ per condition) supports directional insights only.

Sample Constraints

01

An extremely limited sample ($N = 6$) severely limits statistical power and the ability to detect true effects across all AOI metrics.

Geographic & Demographic

02

The sample was restricted to the Chicago urban area, so psychographic and dining behaviors cannot be generalized to broader populations.

Laboratory vs. Real-World

03

A rigid desktop / iMotions setup fails to capture the realistic, multi-screen, distracted environment in which consumers actually view ads.

ONE-PAGE SUMMARY

Key Takeaways



BOTTOM LINE

Emotion wins the first glance; product wins the memory. The strongest ad does both.

H1 PARTIALLY SUPPORTED

Emotional headline drove 75% more dwell

"FEELING LOW?" held the eye far longer (1,700 vs 972 ms) - the only statistically significant result ($p = .035$).

THE TRADE-OFF

But emotion crowded out the product

Version B scored 0% Big Mac recall and weaker brand, social-proof, logo and CTA attention - the Vampire Effect.

WHAT WORKS

Product-forward layouts drive memory

Version A led on recall (67%), logo noticeability (100%), social proof (6.3 fixations) and purchase intent.

RECOMMENDATION

Build the hybrid - and re-test bigger

Pair B's emotional headline with A's product-forward layout; validate at $N \geq 30$ on mobile before rollout.

IF WE RAN THE STUDY AGAIN

Future Directions

STUDY IMPROVEMENTS

- Increase sample size ($N \geq 30$)
- Add diverse cities, ages & cultures
- Test on mobile & social platforms
- Add facial coding / EEG signals
- Balance emotional vs product stimuli

Hybrid Ad Testing

Test a hybrid ad combining B's emotional headline with A's product-focused layout to capture attention and recall.

CTA Placement

Experiment with CTA placement and product positioning to optimize gaze flow and lift conversion.

Social-Proof Impact

Measure how ratings, reviews and "#1 Choice" badges affect visual attention and purchase intent.

Static vs Video

Compare static ads with short-form video on TikTok, Instagram and YouTube for engagement.

Menu Simplicity

Test menu simplicity and visual hierarchy to reduce cognitive load and improve intent.

Platform-Specific

Run ads on mobile and social platforms to reflect real-world, multi-screen viewing.

GROUP 7 · MKT 546 NEUROMARKETING · DEPAUL UNIVERSITY

Thank You!

Q&A - QUESTIONS?

Francesca De Martino

Disha Surati

Kristal Nagre

Khushi Shaikh

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Naveen Raj Kanagaraj

Appendix



Donald's

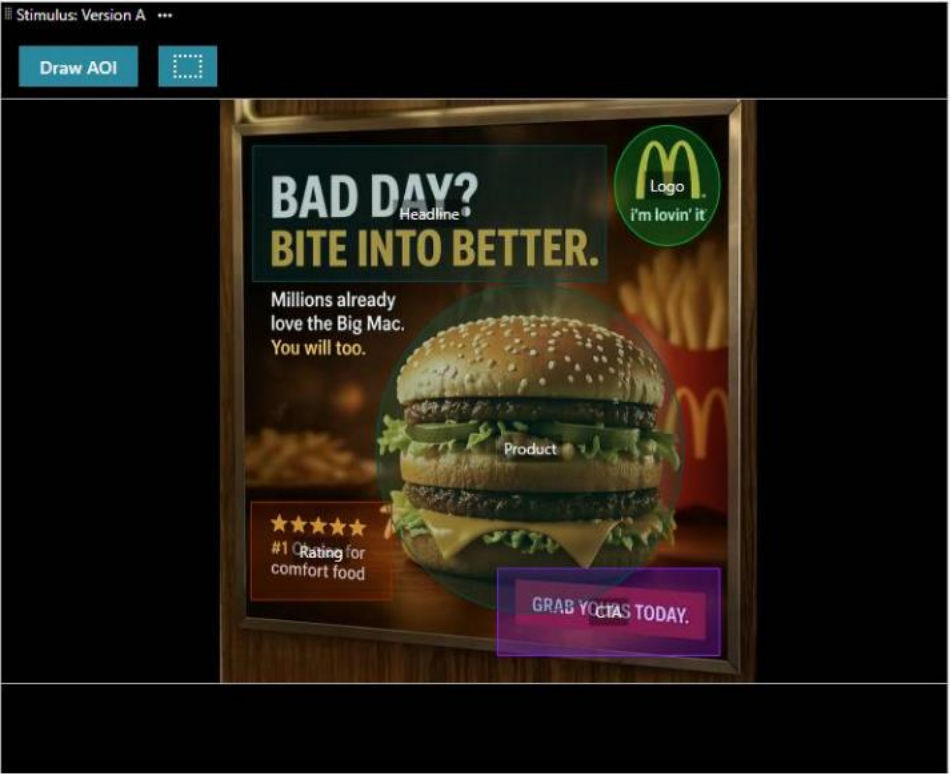


A

Stimulus A

AOI Metrics

Eye tracking AOI metrics for Version A (Product-Focused). Fixation counts, dwell times, and time to first fixation per area of interest.



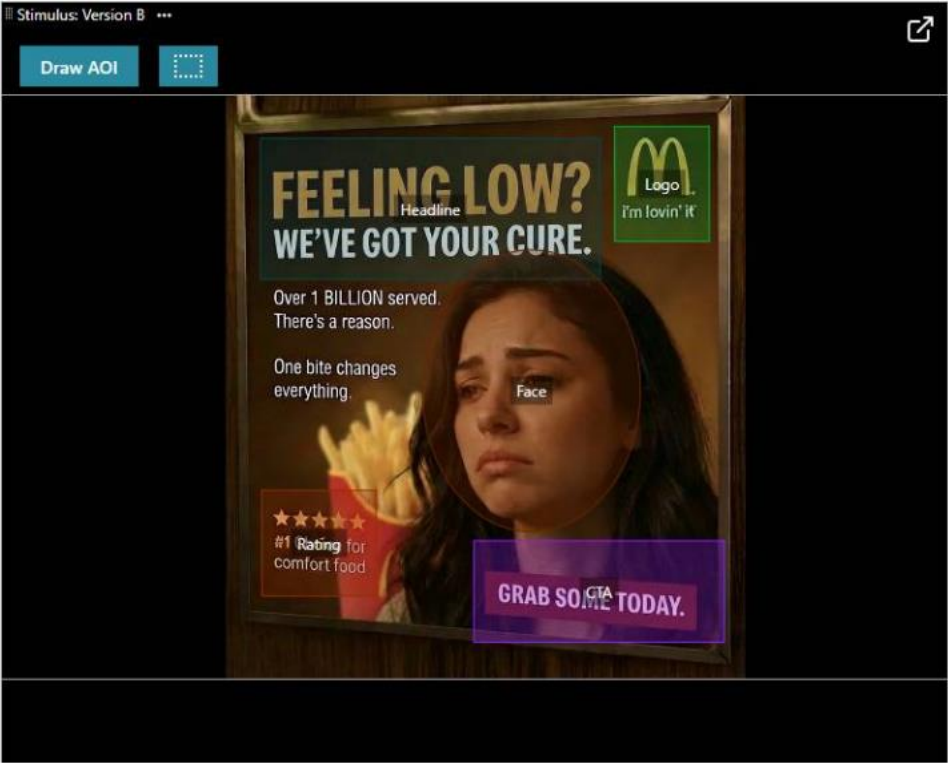
AOI metrics	Logo	CTA	Product	Headline	Rating
Information					
AOI duration (ms)	9992.3	9992.3	9992.3	9992.3	9992.3
AOI duration (%)	100	100	100	100	100
Size (cm2)	25.2	49.2	202.4	122	35.1
Size (%)	1.6	3.2	13	7.8	2.3
Respondent base	3	3	3	3	3
Fixation based metrics					
Respondent ratio (%)	100	66.7	100	100	100
Revisit count	1	1	3.7	0.7	1
Fixation count	4	4	6.7	6	6.3
TTFF AOI (ms)	2985.4	4415.1	75.3	535.3	5196.6
Dwell time (ms)	1122.3	675	1510	971.6	1683.6
Dwell time (%)	11.2	6.8	15.1	9.7	16.9
First fixation duration (ms)	494.5	191.7	243.3	172.2	522.2
Saccade based metrics					
Respondent ratio (%)	100	66.7	66.7	100	100
Saccade count	2	2.5	3	6.3	4.3
Amplitude (deg)	1.6	2.5	3.5	2.8	1.4
Peak velocity (deg/s)	51.2	70.8	87.8	79.1	46.4

B

Stimulus B

AOI Metrics

Eye tracking AOI metrics for Version B (Emotion-Focused). Fixation counts, dwell times, and time to first fixation per area of interest.

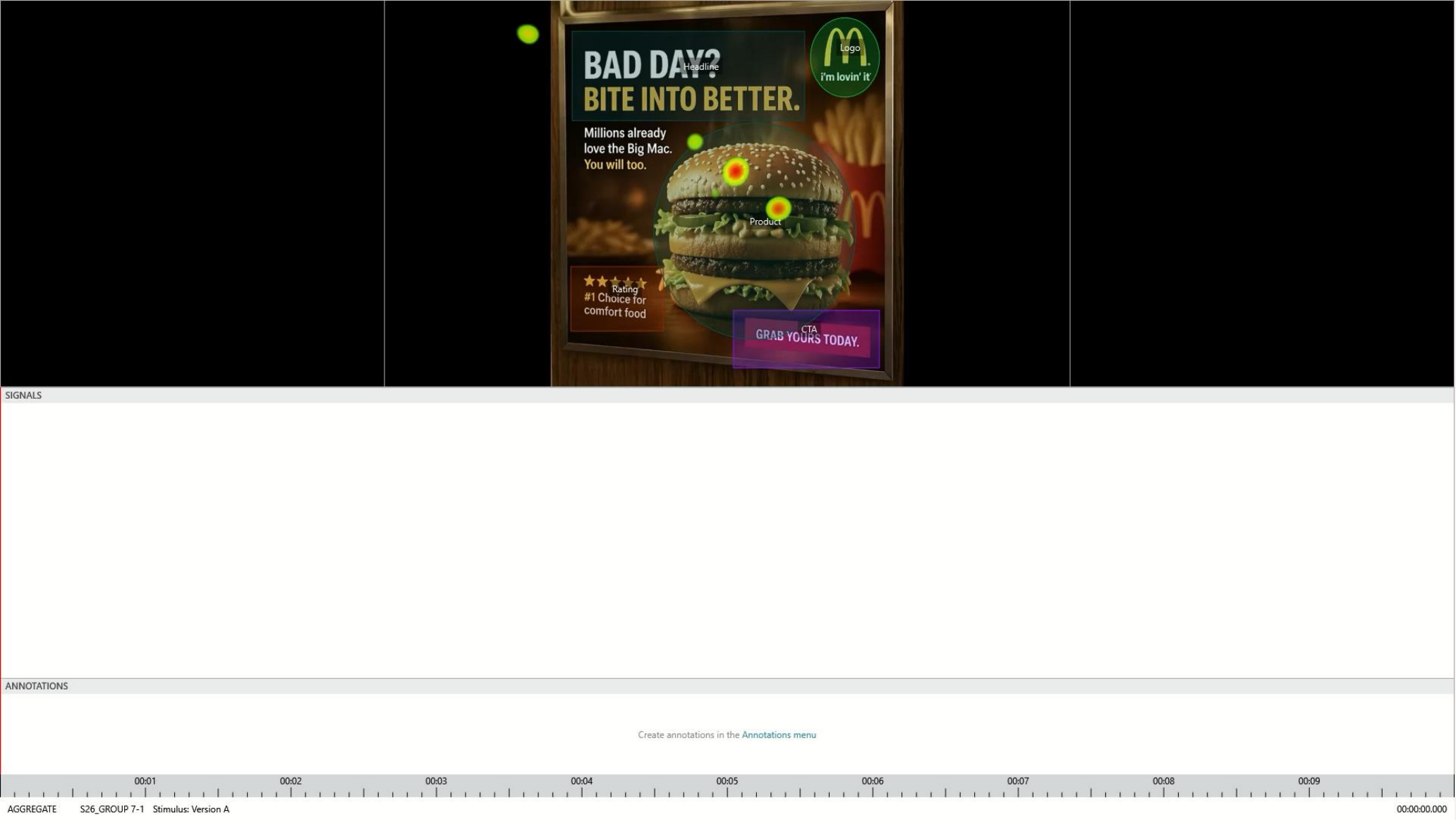


AOI metrics	Headline	Logo	Face	CTA	Rating
Information					
AOI duration (ms)	9994.7	9994.7	9994.7	9994.7	9994.7
AOI duration (%)	100	100	100	100	100
Size (cm2)	122.6	27.8	123.5	64.9	31
Size (%)	7.9	1.8	7.9	4.2	2
Respondent base	3	3	3	3	3
Fixation based metrics					
Respondent ratio (%)	100	66.7	100	66.7	66.7
Revisit count	2.3	0.5	3.3	0	0
Fixation count	10.3	3	8	2	1
TTFF AOI (ms)	559.5	4852.2	0	6018.9	5443.9
Dwell time (ms)	1700.1	1133.4	1591.7	383.4	508.4
Dwell time (%)	17	11.4	15.9	3.8	5.1
First fixation duration (ms)	150	233.3	265	258.4	508.4
Saccade based metrics					
Respondent ratio (%)	100	66.7	100	100	0
Saccade count	18.3	1.5	7.3	1	NA
Amplitude (deg)	2.7	0.9	2	1.5	NA
Peak velocity (deg/s)	81.4	43.3	72.1	57.5	NA

C

Bee Swarm A

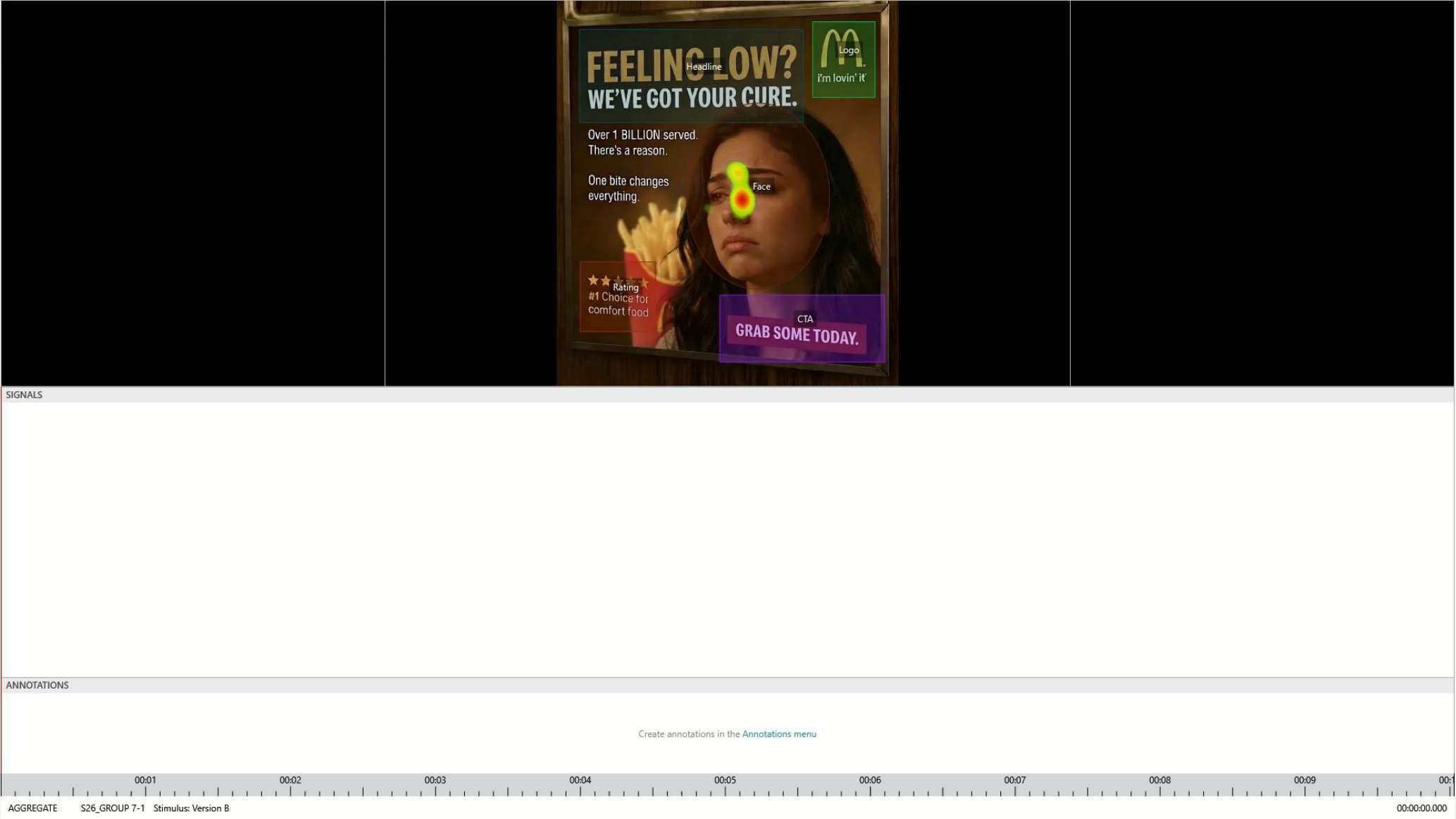
Gaze point distribution (bee swarm) for Version A - Product-Focused stimulus. Each dot represents a single fixation event.



D

Bee Swarm B

Gaze point distribution (bee swarm) for Version B - Emotion-Focused stimulus. Note concentration on face AOI.



E

ANOVA Test

One-Way ANOVA statistical test results.
Full output including F-statistics and p-values for all measured metrics.

ANOVA results

One-Way ANOVA +
Mann-Whitney U

Why ANOVA? One-Way ANOVA compares means across two independent groups (Version A vs. B). Mann-Whitney U used as non-parametric check - n=3 per cell means normality cannot be verified. Results are exploratory.

Measure	Version A	Version B	F-statistic	p-value	Result
Headline Fixation Dwell (ms)	M = 972	M = 1,700	F(1,4) = 9.95	p = 0.035 ★	SIGNIFICANT
Social Proof Dwell (ms)	M = 1,684	M = 508	F(1,3) = 4.06	p = 0.137	Large effect, n.s.
Purchase Intent (1–5 scale)	M = 2.0	M = 1.3	F(1,4) = 0.40	p = 0.563	Not significant
Ad Clarity (1–10 scale)	M = 6.0	M = 4.7	F(1,4) = 0.24	p = 0.651	Not significant

★ Only Headline Fixation Dwell Time reaches statistical significance. All other results are directional only - N ≥ 15 per cell is required for inferential conclusions.

F

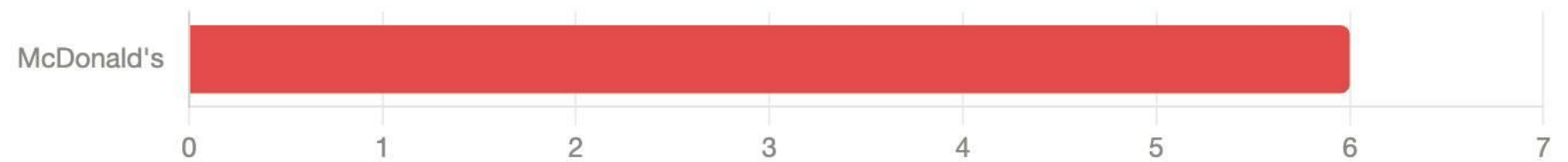
Brand Recognition

Post-exposure brand recognition survey results. Measures unaided and aided recall of McDonald's brand elements.

Q2 · Brand recognition

What brand was shown in the advertisement?

■ McDonald's — 6



G

Headline Recall

Post-exposure headline recall data.
Compares recall rates for Version A ('Feel the Crunch') and Version B ('Feeling Low?') headlines.

Q3 · Headline recall (open text)

What was the main headline or message of the ad?

"Feeling Low?"

"Bite into This"

"Billion was all I saw"

"Something about first bite"

Don't remember (×2)

H

Product Identification

Product identification accuracy rates - correctly identifying the Big Mac after ad exposure, by stimulus version.

Q4 · Product identification

What product was featured in the ad?

■ Burger / Big Mac — 4 ■ Fries / French Fries — 2



Ad Tone

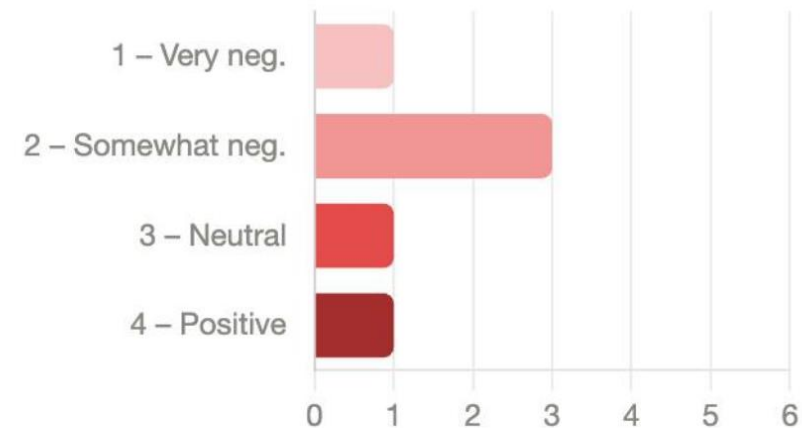
Survey responses on overall ad tone (1–4 scale) and ad clarity (1–10) across both ad versions.

Q5 · Ad tone

Overall tone (1–4 scale)

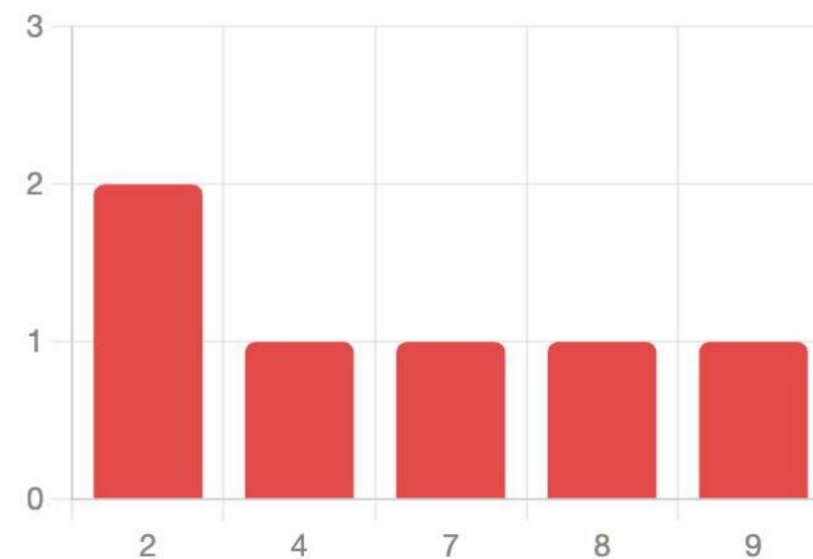
1 = Very neg. 2 = Somewhat neg. 3 = Neutral

4 = Positive



Q6 · Ad clarity

Clarity rating (1–10)



J

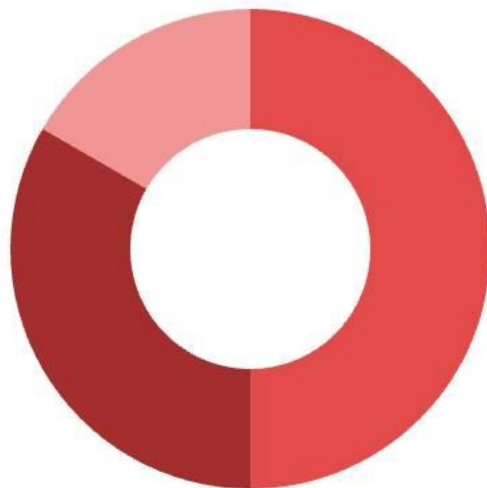
Demographics

Full demographic breakdown of the study sample. Age, gender, education, and fast food consumption frequency.

Q7 · Demographics

How do you identify your gender?

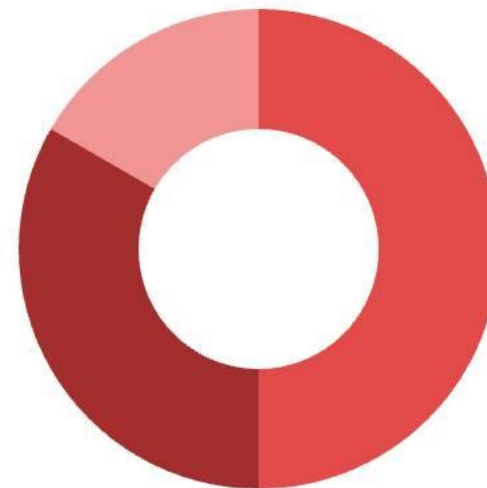
- Female (2) — 3
- Male (1) — 2
- Non-binary (3) — 1



Q8 · Demographics

What is your age range?

- 18-24 (2) — 3
- 25-34 (3) — 2
- Under 18 (1) — 1



K

Purchase Behaviour

Purchase behavior patterns including fast food frequency, McDonald's purchase recency, and spending habits.

Q9 · Purchase behaviour

After viewing this ad, how likely are you to purchase the advertised product?

■ Not likely (1) — 4 ■ Somewhat unlikely (2) — 1 ■ Likely (4) — 1



Behaviour

General consumer behavior patterns including media consumption, dining preferences, and convenience orientation.

Q10 · Behaviour

How often do you eat fast food per week?

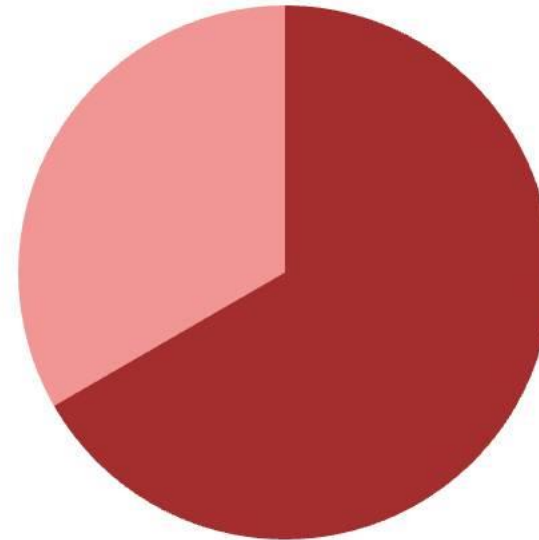
■ 4-5 times (5) — 3 ■ Less than once (1) — 2 ■ 3-4 times (4) — 1



Q11 · Recency

Have you purchased a McDonald's product in the last 30 days?

■ No (2) — 4 ■ Yes (1) — 2



N

Purchase Intent

Purchase intent ratings (1-5 scale) after exposure to each stimulus. Version A: M=2.0, Version B: M=1.3 (not significant, $p=0.563$).

Q12 · Intent

How likely are you to purchase fast food within the next week?

■ Likely (4) — 3 ■ Somewhat unlikely (2) — 2 ■ Not likely (1) — 1



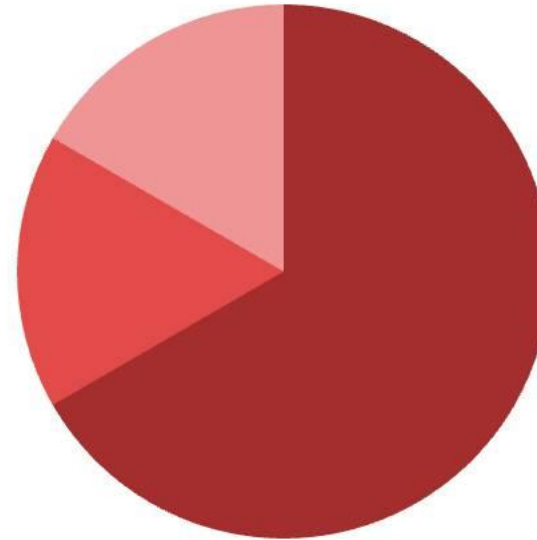
O Media

Media consumption channel data. Social media was the most common advertising channel among respondents.

Q13 · Media

Where do you typically encounter fast food advertising?

■ Social media (1) — 4 ■ TV (3) — 1 ■ Online ads (2) — 1

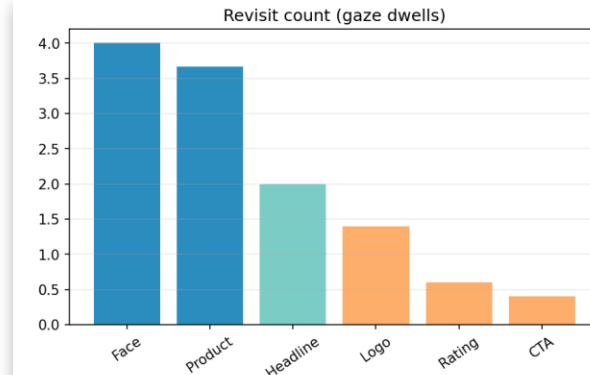
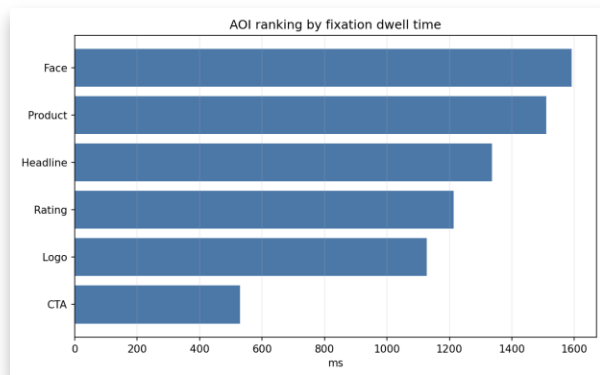
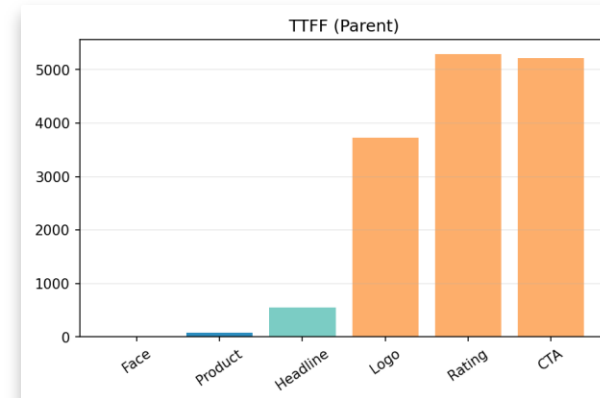
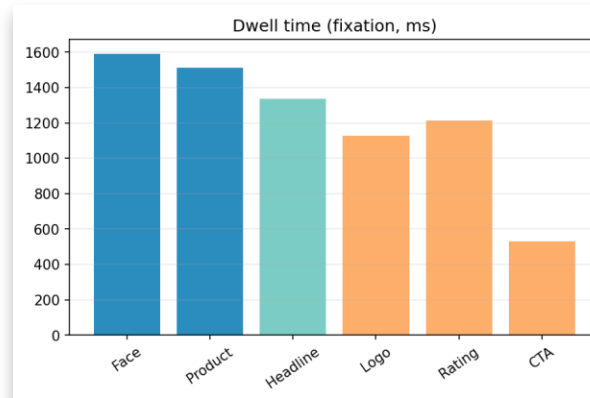
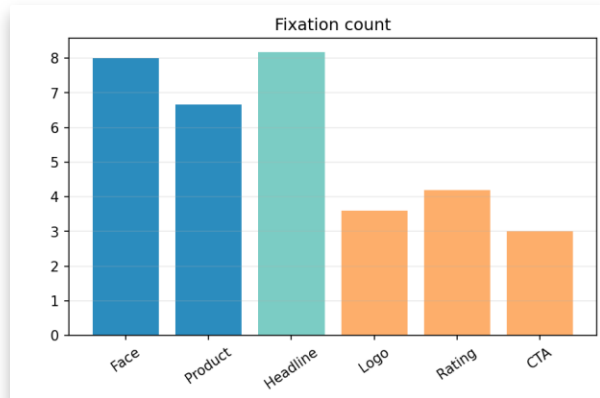


APPENDIX P

P

AOI Metrics

Complete chart set for all Areas of Interest metrics across both ad versions.



KEY FINDING

Only Headline Fixation Dwell Time reached statistical significance

$$F(1,4) = 9.95, p = 0.035$$

Face and Product AOIs draw the most attention across fixation count and dwell time.